

2022 Ph H2 Q5

Section: Particles and Waves / Our Dynamic Universe

Topic: Doppler Effect, Redshift, Gravitation

A buzzer swung in a circle demonstrates the Doppler effect. (a) Explain frequency change using wavefronts. (b) Binary star spectra: (i) calculate redshift, (ii) estimate distance. (c) Gravitational force between stars: (i) calculate force, (ii) effect of halving separation.

Worked solution

(a) Doppler effect with wavefronts

- When the buzzer moves towards the students, successive wavefronts are emitted closer together. The wavelength is shortened and frequency appears higher.
- When the buzzer moves away, wavefronts are emitted further apart, wavelength increases and frequency is lower.

(b)(i) Redshift

$$\begin{aligned} z &= (\lambda_{\text{obs}} - \lambda_{\text{rest}}) / \lambda_{\text{rest}} \\ &= (610 - 580) / 580 = 0.052 \end{aligned}$$

Answer: $z \approx 0.052$

(b)(ii) Distance using Hubble's law

$$v \approx zc = 0.052 \times 3.0 \times 10^8 = 1.56 \times 10^7 \text{ m s}^{-1}$$

$$d = v / H_0 = (1.56 \times 10^7) / (2.3 \times 10^{-18}) = 6.75 \times 10^{24} \text{ m}$$

Convert to light years: $\approx 2.2 \times 10^9 \text{ ly}$.

Answer: $\approx 2 \times 10^9$ light years

(c)(i) Gravitational force

$$F = G m_1 m_2 / r^2$$

$$= (6.67 \times 10^{-11})(2.19 \times 10^{30})(1.80 \times 10^{30}) / (3.44 \times 10^{12})^2$$
$$= 2.22 \times 10^{25} \text{ N}$$

Answer: $\approx 2.2 \times 10^{25} \text{ N}$

(c)(ii) Effect of halving separation

Force $\propto 1/r^2$. If $r \rightarrow r/2$, F increases by factor $(2^2) = 4$.

Answer: 4 times greater

Final answers

(a) Towards: wavefronts closer, higher f . Away: wavefronts spread, lower f .

(b)(i) Redshift $z \approx 0.052$

(b)(ii) Distance $\approx 2 \times 10^9 \text{ ly}$

(c)(i) $F \approx 2.2 \times 10^{25} \text{ N}$

(c)(ii) 4 times greater

Revision tips

- Doppler effect: moving source compresses/expands wavefronts.
- Redshift $z = \Delta\lambda / \lambda_0$.
- Hubble's law: $v = H_0 d$ links redshift to distance.
- Newton's law of gravitation: $F \propto 1/r^2$.
- If distance halves, gravitational force $\times 4$.